

WHITE PAPER

Domestic Thermal Energy Store

A white paper on Hestora: whole-home heating and hot water from cheap, clean, off-peak electricity — in the homes we already have.

Hestora Limited · August 2026 · www.hestora.net



Hestora occupies the space of a hot-water cylinder — here in a utility room, but it installs anywhere in the home: garage, airing cupboard or loft.

1. In brief

The UK has to take fossil-fuel boilers out of around 25 million homes, and the only energy source that scales to replace them is electricity. At the same time, the electricity system increasingly produces very cheap, very clean power overnight — power it sometimes pays wind farms to throw away. **Hestora is built to sit precisely in the gap between those two facts.**

Hestora is a 40 kWh pressurised, water-based thermal energy store. It charges from cheap off-peak electricity in a four-to-five-hour overnight window, then delivers a full day of space heating and mains-pressure hot water through the radiators and pipework a home already has. Installation takes around a day and it occupies the space of a hot-water cylinder — 650 mm × 650 mm and 1,800 mm tall — with no outdoor unit, no new radiators, and no change to how the household experiences heating. Because the unit arrives light and dry, it can be sited almost anywhere in the home: garage, utility room, airing cupboard or loft.

One device, three aligned benefits

Consumer — shifts the household's largest energy load into the cheapest, cleanest hours, cutting and stabilising running costs while keeping a boiler-like experience.

Grid — a controllable demand asset that can absorb 40 kWh on command, moving demand away from the peaks and providing the flexibility a renewables-heavy system needs.

Renewables — gives surplus overnight wind and solar somewhere useful to go, converting would-be curtailment into stored domestic heat.

2. The problem: a timing mismatch

Domestic heating and hot water account for roughly a fifth of UK carbon emissions, and around 85% of homes still burn natural gas. Electricity generation has decarbonised quickly; heat has barely moved. The structural difficulty is timing: renewable output is weather-driven, while heat demand peaks on winter mornings and evenings — exactly when electricity is most expensive and most carbon-intensive.

The cost is no longer theoretical. In 2025 Great Britain paid almost **£1.5 billion in constraint costs** — roughly £380m paying wind farms to switch off and over £1bn firing up gas plant to replace the lost output — recovered through every household's bill. NESO's own analysis suggests balancing costs could reach £4–8 billion a year by 2030. Electrifying heat without storage makes this worse: every heat pump and panel heater that lacks storage adds load to the evening peak the grid can least afford to serve.



Two sides of the transition on one street: homes still burning gas opposite new-builds already generating their own solar power.

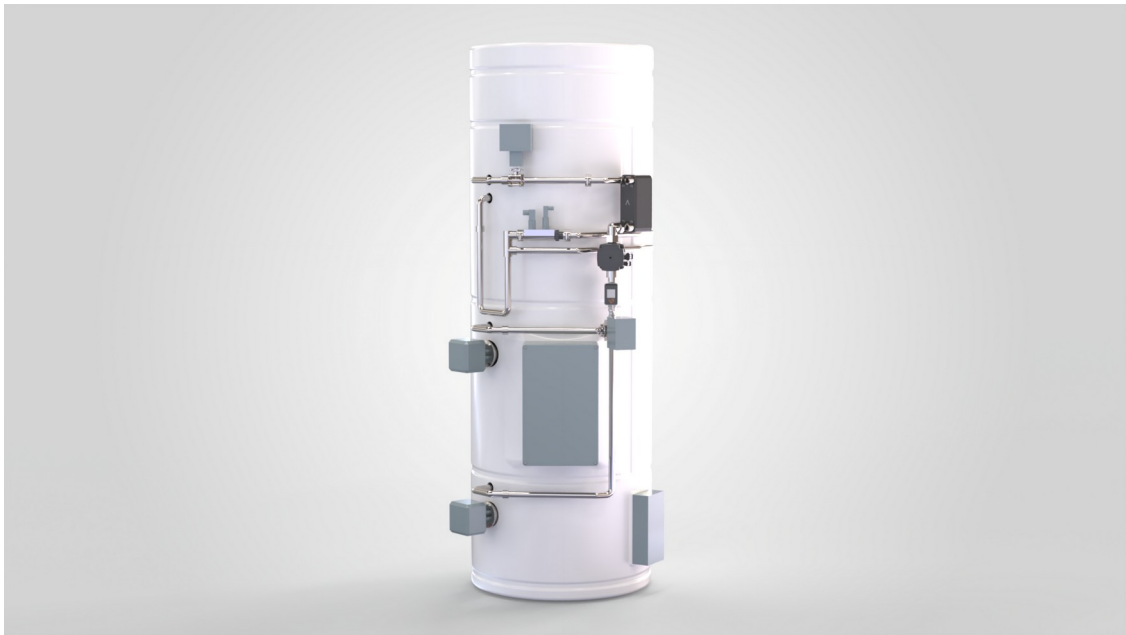
3. How Hestora works

Hestora is a domestic heat store. It charges itself with cheap electricity during the off-peak window, holds a full day's heat, and releases it on demand — as mains-pressure domestic hot water and to a conventional wet central-heating circuit. From the occupant's

side it behaves exactly like a boiler: turn on the tap, turn up the thermostat, and the heat is there — without the emissions or the exposure to fossil-fuel price shocks.

What makes that possible is pressurised water. Held under pressure in a sealed vessel — up to 8 bar — water stays liquid well above its normal boiling point, and Hestora stores it at up to 175 °C. That is what allows a whole day's demand to fit in the space of a hot-water cylinder: about 2.4 times the energy of a conventional cylinder of the same volume held at 65 °C. Heat is delivered on demand through heat exchangers, and water is free, non-toxic and non-degrading — which is why the store is expected to outlast the house it is fitted in.

- **Water as the storage medium** — free, non-toxic, non-degrading and thermodynamically predictable. No phase-change materials, no lithium cells, no refrigerants.
- **Filled on site** — the unit arrives light and dry and is filled with tap water once in position, so it can go through hallways, up stairs and into lofts. Solid-core thermal stores of comparable capacity arrive as a single mass of 300 kg or more, which in practice confines them to ground-floor locations; Hestora can go where a hot-water cylinder goes.
- **Flexibility without behaviour change** — the control system is being developed to interpret smart-grid signals and time-of-use tariffs and decide when to charge. The household will neither notice nor manage it — which is what makes the flexibility scalable.



The Hestora MKII: pressure vessel, high-performance insulation, heat exchangers and smart controls in one appliance.

Specification

Specification	Value	Status
Energy capacity / heat output	40 kWh / 30 kW	✓ Full-size prototype
Operating range	50-175 °C / 0-8 bar g	✓ Verified
Vessel certification	PD 5500:2024+A2 Category 2	✓ Independently certified
Round-trip efficiency	~92% (~4 kWh/day standing loss, largely recovered as space heat)	🕒 In-house complete; pilot 2026

Specification	Value	Status
Solar integration	Built-in diverter — charges the store from the array ahead of the grid	● Confirmed; pilot 2026
Design life	>30 years	— Engineering target
Install	Around a day; dry install, filled in situ	✓ Demonstrated
Dimensions / siting	650 × 650 × 1,800 mm — the space of a hot-water cylinder; garage, utility, airing cupboard or loft	✓ Demonstrated

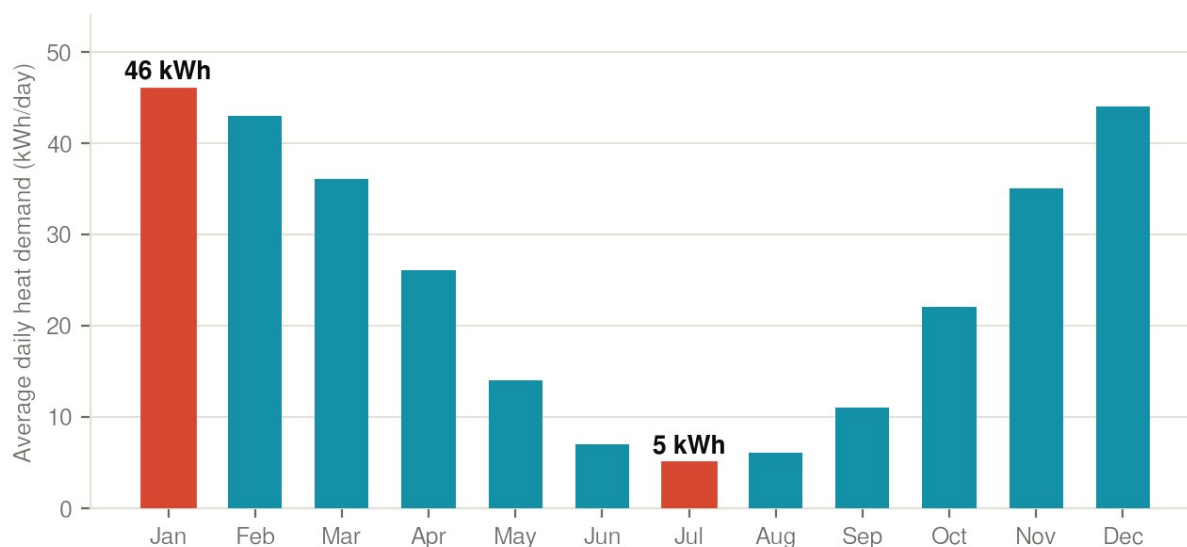
✓ verified or independently certified · ● validation in progress · — engineering target, not yet tested. Vessel hydraulically tested to 17.86 bar g against a maximum working pressure of 8 bar g — a factor of 2.23 — by British Engineering Services. Standing loss ~4 kWh/day, largely recovered as useful space heat within the dwelling.

4. Sized from 6.4 million real meter readings, not assumptions

Hestora's 40 kWh capacity is grounded in primary data: a living-lab analysis of a full year of half-hourly smart-meter readings — about 6.4 million data points — from 200 British homes. The typical gas-heated home uses about 8,950 kWh a year (79% space heating, 18% hot water, 3% cooking), collapsing to around 5 kWh a day in July and peaking near 66 kWh on the coldest January days.

A note on comparison with published figures: Ofgem's typical domestic consumption value for a medium gas-heated household is 9,500 kWh a year (revised 1 July 2026, down from 11,500 kWh). That figure is gas purchased; ours is heat delivered. At a typical 85% seasonal boiler efficiency, 9,500 kWh of gas yields about 8,075 kWh of useful heat — about 10% below the 8,950 kWh we measure, part of which is the 3% cooking component a boiler-efficiency conversion does not capture. The sizing case therefore rests on slightly higher demand than the current TDCV implies, not lower.

Heat demand of a typical UK gas-heated home across the year



Average daily space-heating and hot-water demand by month for a typical UK gas-heated home ($\approx 8,950$ kWh/yr). January demand runs at roughly nine times July's — which is why storage must be sized to a genuine winter day, not an annual average. Source: Hestora living-lab dataset — a full year of half-hourly smart-meter data from 200 British homes, ~ 6.4 million readings.

Why 40 kWh?

A single 40 kWh store covers the daily heat demand of roughly **half of UK homes** — 47% of the stock has a maximum daily demand below 45 kWh — so whole-home coverage is a measured claim, not a brochure one. On the deepest winter days the control system simply tops up in additional cheap windows during the day.

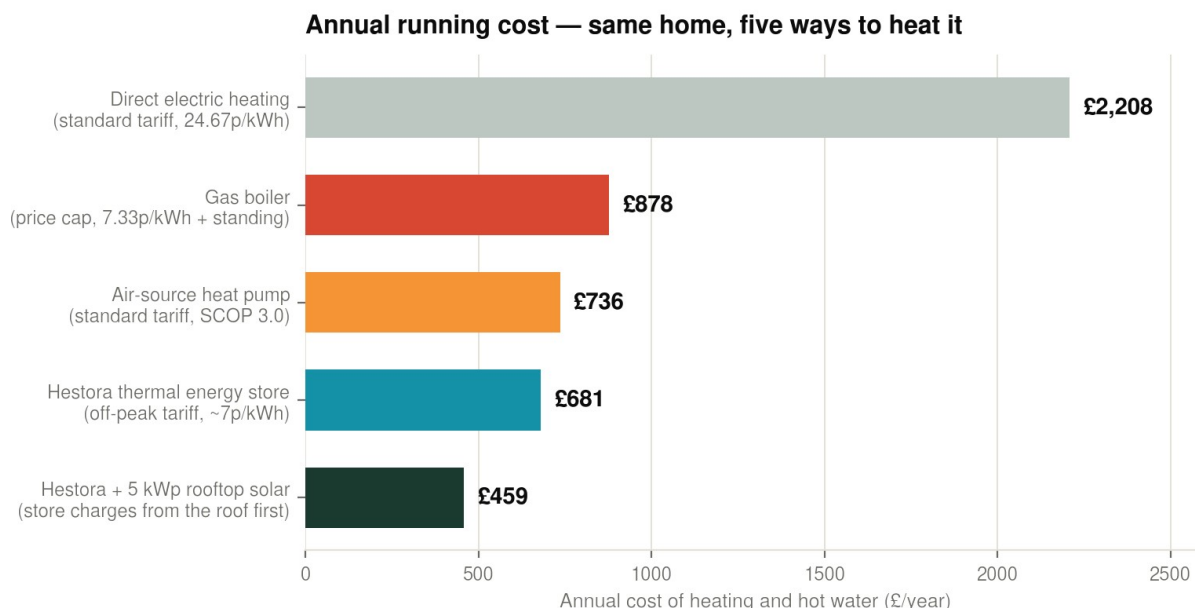
Capacity is a bills argument, not a spec-sheet one.

The size of the store decides how much of the year's heat can be bought at the cheap rate. A store that runs dry on a cold evening makes up the difference at the daytime price — three to four times as much. At 40 kWh, Hestora covers a genuine winter day for roughly half the housing stock, so for those homes almost all of the year's heat is bought inside the off-peak window.

Output matters too. Hestora delivers up to 30 kW — more than most domestic boilers, and more than any heat pump sized for a house of this type. There is no recovery wait and no trade-off: a bath and the central heating can run at once.

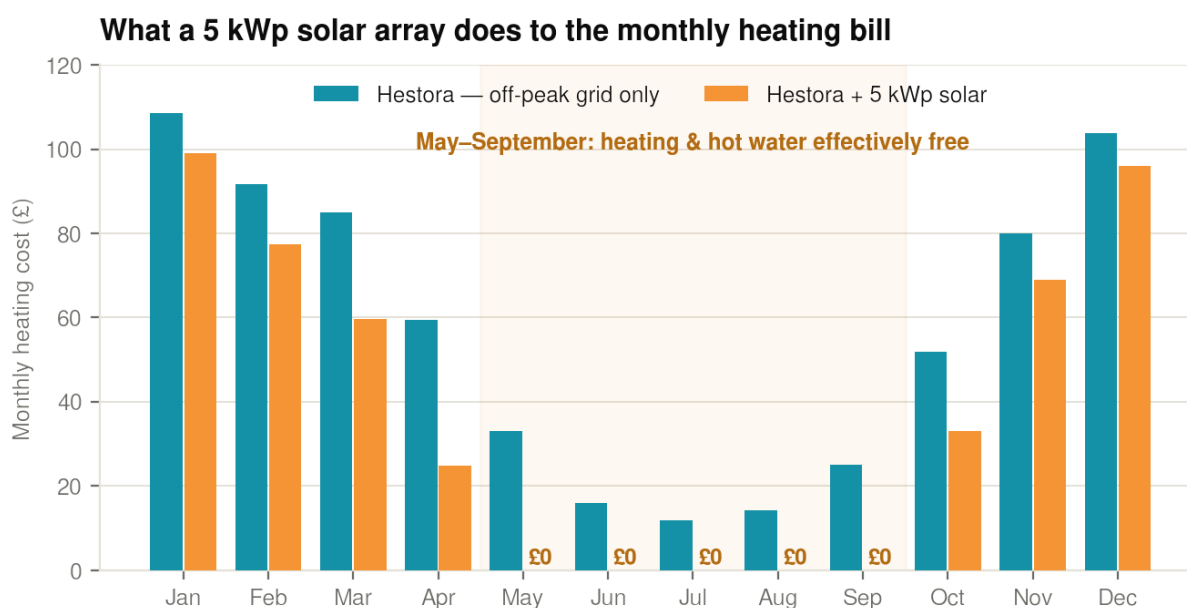
5. What it costs to run

Heating is the largest component of most households' energy use, so shifting it matters disproportionately. Hestora moves that load into the overnight window, where smart time-of-use tariffs now price electricity at around 6.7–7p/kWh — a quarter of the standard daytime rate.



Modelled annual cost of meeting 8,950 kWh of heat and hot-water demand, mid-2026 prices. Assumptions: direct electric at the standard single rate (24.67p/kWh); gas at the Jul-Sep 2026 cap (7.33p/kWh plus £106/yr standing charge, 85% boiler efficiency); air-source heat pump at SCOP 3.0 on the standard rate, spanning £631–£788 across the plausible 3.5–2.8 range; Hestora at ~7p/kWh off-peak with 92% round-trip efficiency. The final bar adds a 5 kWp array with the store charging from the roof first. An all-electric home also sheds the gas standing charge.

Rooftop solar makes the picture stronger still. A built-in diverter charges the store from the home's own array before anything is bought from the grid — and depending on the size of the array, that can mean **virtually no heating and hot-water costs from late spring to early autumn**, with the store soaking up the middle-of-the-day generation that would otherwise be exported for pennies.



Monthly heating and hot-water cost with Hestora charged from off-peak grid electricity alone (blue) and with a 5 kWp south-facing array feeding the store ahead of the grid (amber). Summer generation exceeds the store's requirement, so May–September falls to zero and the annual cost drops from ~£681 to ~£459 — the array supplies roughly a third of annual charging energy. Monthly-resolution model against the living-lab demand profile; a larger array widens the zero-cost window.

Just as important is stability: by decoupling consumption from real-time prices, the household is insulated from winter-evening price spikes.

The carbon position is better than the cost position suggests, and it moves with the season. Through the depth of the heating season — October to February, which carries 57% of a home's annual heat demand — the overnight window is the cleanest period of the day, running 12–34% cleaner on monthly averages between midnight and 5 a.m. than at midday. From March onwards that reverses, because the cleanest hours shift to the middle of the day when solar is generating, and Hestora's diverter charges the store from the roof instead. The two mechanisms cover opposite halves of the year, which is why the store is designed to use both.

Measured across a full year of NESO actuals and weighted by when heat is actually consumed, charging overnight rather than at the average grid intensity reduces the carbon intensity of Hestora's heat by 14% — from 125 to 108 gCO₂/kWh. That figure also falls as the system decarbonises: the same overnight window ran at 149 gCO₂/kWh three years ago against 108 today. The overnight advantage itself has held steady at 14–15% across those three years, so the improvement is one any electrified home shares; what storage adds is the shift into the cleaner window in the first place.

Calculated from NESO Carbon Intensity API actuals (api.carbonintensity.org.uk), 1 August 2025 to 31 July 2026 — 17,520 half-hourly periods with no missing data. Windows are local clock time. Weighted by monthly heat demand derived from an England degree-day profile plus a flat hot-water component (71%/29% by annual energy). Overnight intensity is weather-driven and varies widely: across the 365 individual days, unweighted, the tenth and ninetieth percentiles were 55 and 184 gCO₂/kWh. Compliance calculations under the Future Homes Standard apply a single fixed factor of 86 gCO₂/kWh, which projects forward across the assessment period; the figures here are measured present-day intensities and the two are not directly comparable. Method and code available on request.

6. Where it sits among the alternatives

Approach	Limitation it carries	Hestora's advantage
Conventional hot-water cylinder	Low energy density; passive; no demand-shifting	$\approx 2.4\times$ the energy in the same volume (cylinder held at 65 °C); whole-home heat and hot water; grid-responsive
Heat pump (+ buffer)	£8,000–£14,000 installed; disruptive retrofit; fabric-dependent; still peak-exposed	Swap in around a day, no outdoor unit, no emitter upgrades, peak-shifting
Phase-change (PCM) storage	Material cost; cycling degradation	Water core: no degradation, low cost, long life
Li-ion battery + heater	3–5 $\times$ cost per kWh of heat; typically warranted to 80% capacity at 10 years; fire considerations	>30-year design life; no cells; no capacity fade
Solid-core electric block	Arrives as a single mass of 300 kg or more, which in practice confines siting to the ground floor	Light dry install, filled in situ; heat and hot water combined

The density difference is structural rather than incremental. A vented or unvented cylinder is not a pressure vessel, so it cannot hold water above its boiling point however well insulated it is — and in practice it is held at around 65 °C, because a poorly insulated tank held hotter simply loses the heat. Hestora is certified to PD 5500 and stores at up to 175 °C. Storing 40 kWh in water at 65 °C would take roughly 625 litres — two and a half times the volume Hestora needs — which will not go in an airing cupboard. Hestora holds it in 265 litres, in the footprint of a single cylinder. (Both figures assume a 10 °C cold feed.)

Heat pumps remain excellent for homes that suit them. Hestora is designed for the homes they struggle to serve — no outdoor space, tight access, conservation constraints, poor fabric, or a tighter budget. Both displace gas; they suit different homes.

7. New developments and network capacity

Electrifying heat across a development raises the coincident peak demand a connection has to be sized for, and where that exceeds the available network capacity the developer funds the reinforcement. Because Hestora charges overnight and draws nothing at the evening peak, each unit reduces the coincident maximum demand rather than adding to it. On a constrained site, storage can be the difference between connecting within existing capacity and paying for an upgrade — a cost that falls on the developer's budget rather than the occupant's bill.

The same property matters under the Future Homes Standard, which from 24 March 2027 requires low-carbon heating in every new home. A development that meets it with unstored electric heating concentrates its new load in the evening peak; one that meets it with storage does not.

8. Status and route to market

- **Certified prototype** — full-size unit built; pressure vessel independently certified to PD 5500:2024+A2 (Category 2) by British Engineering Services and hydraulically

tested to more than twice its maximum working pressure — 17.86 bar g against 8 bar g, a factor of 2.23.

- **Instrumented pilot** — a monitored pilot begins in November 2026.
- **Certification & grants** — access to the £2,500 Boiler Upgrade Scheme grant depends on the BSI standard for thermal energy storage being defined, alongside MCS certification and testing by BRE. Like other thermal energy stores, Hestora is waiting for these to be confirmed before the funding opens; once it does, the grant plus ~£1,000 of carbon-offset funding would bring the net customer cost of a £7,000 install towards £3,500.
- **Commercial sales** — planned to open in 2027, delivered through a trained installer network with an optional service subscription covering tariff optimisation, monitoring and warranty.
- **Grid revenue** — Phase 1 charging uses the overnight off-peak window. From mid-2027, units will additionally integrate with NESO's Demand Flexibility Service 'turn-up' mechanism — which since April 2026 pays controllable demand to absorb surplus renewable power. Coupling off-peak charging with paid turn-up events will be a powerful combination: each Hestora can absorb 40 kWh on instruction.

9. About Hestora

Hestora Limited is a British engineering company based in the Midlands. We source components from UK companies to support British engineering, and we have been selected for energy and enterprise programmes at Imperial College London and the University of Birmingham. Our mission is to make zero-emission heating accessible for every home.

Find out more

Visit www.hestora.net to see the product in action, read common questions, and register your interest in the pilot programme. Contact: jim@hestora.net